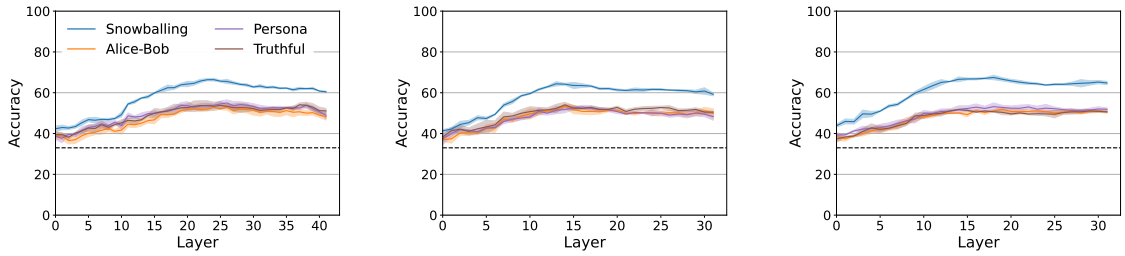
Model	Dataset	HK^+ mitigation (%)	HK^- mitigation (%)
Gemma	TriviaQA	**17.2 / 21.4 / 23.4 / 9.0**	1.8 / 2.2 / 1.8 / 2.6
	Natural Questions	**20.4 / 27.4 / 28.0 / 14.6**	2.4 / 2.4 / 2.4 / 2.0
Llama	TriviaQA	**19.4 / 27.0 / 16.6 / 16.6**	1.6 / 1.6 / 1.0 / 2.0
	Natural Questions	**16.6 / 24.4 / 25.2 / 21.0**	2.0 / 1.6 / 1.2 / 1.2
Mistral	TriviaQA	**27.4 / 27.8 / 20.0 / 49.2**	1.6 / 3.2 / 0.8 / 2.8
	Natural Questions	**34.2 / 35.2 / 23.0 / 51.0**	2.4 / 3.0 / 1.4 / 2.8

Table 3: Comparison of HK^+ and HK^- mitigation success percentage results on Truthful / Persona / Alice-Bob / Snowballing settings. Mitigating HK^+ is significantly more successful than mitigating HK^-.



(a) Gemma. (b) Llama. (c) Mistral.

Figure 2: 3-way classification results on TrivaQA into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples. All the detection results are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.

query/gold-answer associations), models fail to employ their knowledge in various contexts.

4 Mitigating and Detecting Different Types of Hallucinations

We first investigate whether mitigation works differently for HK^+ and HK^- hallucinations. Then, we examine whether the different types of hallucinations are represented differently inside models.

4.1 Prompt Mitigation Only Works for HK^+

To demonstrate the necessity of distinguishing between HK^- and HK^+ hallucinations, we show that each type requires different mitigation techniques. In particular, HK^-, caused by lack of knowledge, cannot be mitigated effectively through simple prompting alone; instead, they require an external knowledge source. Importantly, our goal is not to comprehensively evaluate mitigation techniques but rather to provide evidence supporting the need to distinguish between HK^+ and HK^- hallucinations, as each type likely requires tailored mitigation methods.

To this end, we prepend the following text before each prompt to encourage the model to generate correct responses (Si et al., 2022; Feng et al., 2024; Taveekitworachai et al., 2024): "Generate answers that are entirely factual and precise, regardless of any issues in the text". (The specific prompt phrasing is not crucial as other prompts work just as well; see Appendix D.)

Results. Table 3 shows the QA accuracy when generating 5 tokens for 500 random examples of both HK^- and HK^+ for all settings.[4] Each cell presents the improvement after using the mitigation.

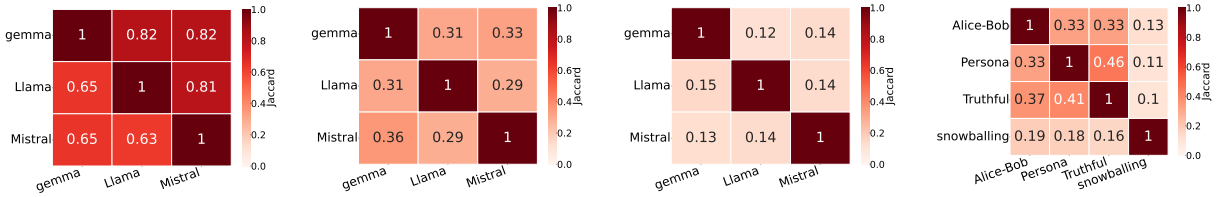
The values for HK^+ are significantly higher than those for HK^- across all settings,[5] underscoring the importance of distinguishing between the two cases for mitigation purposes.

4.2 Simple Classifiers can Moderately Distinguish HK^+ from HK^- from Factually Correct

We next explore whether a classifier can distinguish between hallucinations arising from a model's lack of knowledge and those that occur even when the model possesses relevant information. Prior work on hallucination detection did not focus on this important aspect (Section 2). This differentiation is crucial for understanding hallucinations' underlying mechanisms and developing targeted detection and mitigation strategies. To address a comprehensive scenario, we train a detection classifier to differentiate between three labels, where the model: (1) knows the information and does not hallucinate ('factually correct'), (2) knows the information but hallucinates (HK^+), and (3) does not know the information and thus hallucinates (HK^-). We detect using the model's inner states to inspect whether the model represents these hallucinations differently. Note that this is a challenging task, as it not only requires

[4] Importantly, we report accuracy when generating a sequence of tokens, which is more realistic than comparing probabilities of correct and incorrect answer choices.

[5] For HK^-, the values are not zero, which we attribute to a wrong categorization of HK^- in a small set of cases



(a) Knowledge similarity between models.

(b) HK^+ similarity under Snowballing.

(c) HK^+ similarity under Persona.

(d) Mistral HK^+ similarity between settings.

Figure 3: High-Knowledge and HK^+ differences on TriviaQA (above the diagonal) and Natural Questions (below the diagonal) between the models and settings. Different models have different knowledge and different HK^+ examples in shared knowledge cases.

distinguishing hallucinations from factually correct responses, but also determining the type of hallucination. This requires an assessment of the model's knowledge.

In all detection experiments, we randomly select 1000 examples from each label for analysis in each dataset and split them to 70%/30% for training/testing.[6] We use a linear (SVM) classifier for detection, as in prior work (Li et al., 2023; CH-Wang et al., 2023).[7] The detection results in the main paper are on hidden states from the residual component (after each Transformer block); see Appendix I for similar results on the MLP and Attention components, and Appendix C for additional implementation details.

Results. Figure 2 shows the detection accuracy results for all models and settings on TriviaQA. (Results on Natural Questions are qualitatively similar; Appendix E.) The accuracy at the highest layer is 60%–70% for snowballing and for the other settings around 50%, across three seeds. As expected, the milder settings have lower accuracy than the Snowballing one, but remain well above the random baseline of 33% (dashed line).These results indicate that the models' inner states contain some information for differentiating between the three cases, and a linear classifier can moderately work.[8] However, more research is needed to better distinguish between these hallucinations to improve mitigation techniques.

5 Analysis of HK^+ Hallucinations

In this section, we analyze HK^+ and demonstrate that different models, as well as different settings within the same model, exhibit varying HK^+ patterns. Next, we show despite this inconsistency, our framework can still be used to investigate this phenomenon, by demonstrating that the different HK^+ settings are generalizable.

5.1 Different Models Have Different Knowledge and Different Hallucinations

To demonstrate the heterogeneity in knowledge and hallucinations across models, we measure the Jaccard similarity (also known as intersection over union) of WACK datasets generated for different models. Jaccard values range from 0 (completely dissimilar) to 1 (perfect overlap). First, we compare in Figure 3a the Jaccard similarity of examples deemed as high-knowledge in two models, following our procedure from Section 3.1. The knowledge similarity for Natural Questions (below the diagonal) is approximately 0.6, indicating significant knowledge divergence between models. For TriviaQA (above the diagonal), models exhibit a higher similarity in knowledge (around 0.8).

Next, we compare the Jaccard similarity of HK^+ examples between any two models, out of the set of examples that both models know. Figures 3b and 3c reveal that HK^+ examples are very different between models: 0.1–0.36 Jaccard scores. This difference suggests each model exhibits unique hallucination patterns. Finally, Figure 3d shows that different settings also exhibit unique hallucination patterns within the same model.[9] However, as shown in Section 5.2 different settings do demonstrate some generalization.

These findings emphasize the importance of model-specific approaches to hallucination detection and mitigation, as both knowledge bases and hallucination patterns vary across models and prompt settings.

5.2 Generalization of HK^+ across Settings

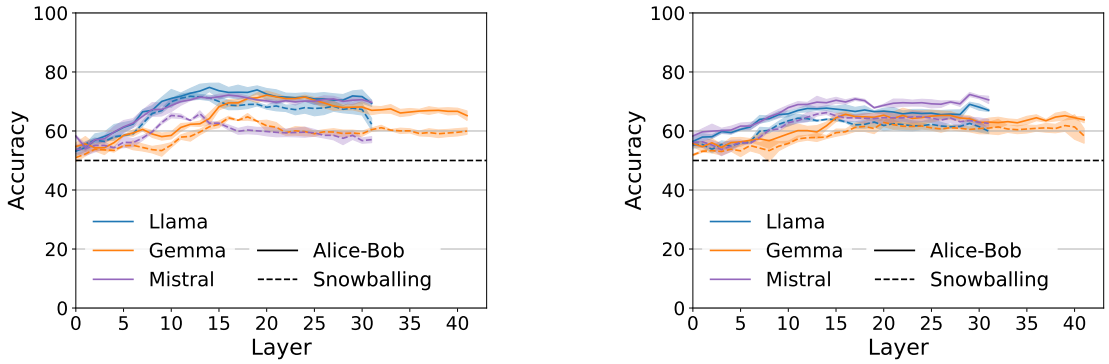
Given that HK^+ hallucinations differ between different prompt settings, how reliable is the choice of the setting? In this section we examine the generalization of hallucination detection across settings. If a classifier trained on one setting generalizes well to another one, this would suggest that the classifier captures a general property of HK^+. Such findings would indicate that these settings are indeed effective for studying the HK^+ phenomenon. Concretely, we evaluate the ability to differentiate between HK^+ and factually-correct examples (binary classification), as these settings only aim at

[6] If there are fewer than 1000 hallucinations we use all the examples we have.

[7] The inputs to the classifier are normalized vectors of the model's inner states at the last answer token.

[8] We observed similar trends in the binary classification case of distinguishing HK^+ from HK^-; Appendix E.

[9] See Appendix F for additional results showing similar patterns across configurations.



(a) TriviaQA.

(b) Natural Questions.

Figure 4: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline.

splitting questions the model knows how to answer into these two categories.

Figure 4 displays the results when testing on the Alice-Bob setting using classifiers trained on either examples from the same setting (solid lines) or from the Snowballing setting (dashed lines). This presents a significant challenge due to the inherent differences between the two settings. While testing on a setting different than the one used for training lowers the results, they are only lower by an average of 5% at the 15th layer and remain above the random baseline, showing some degree of generalization. (See Appendix J for similar results of generalization between other settings.) These positive results lend support to the use of such synthetic datasets for studying the phenomenon of hallucinations despite knowledge.

6 Importance of Model-specific Dataset

After demonstrating that HK^+ is a broad phenomenon and highlighting its uniqueness across different models, we now show that it can benefit from model-specific datasets. First, we compare HK^+ detection using model-specific vs. generic datasets. Next, we show the possibility of detecting HK^+ prior to generation. We argue that these results highlight the importance of model-specific datasets for HK^+.

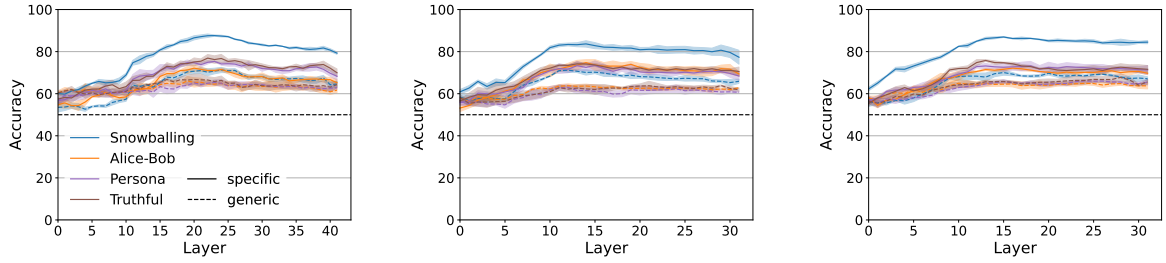
6.1 HK^+ Detections Improves with a Model-specific Dataset

In this section, we show the importance of working with a model-specific dataset instead of a generic one. We start by explaining how to construct the generic dataset and then move to the experiments.

Generic Dataset. A generic dataset is a labeled dataset that does not account for model-specific hallucinations or knowledge. Using a generic dataset is a common practice in the field of hallucination research for both detection and mitigation (e.g., Li et al. (2023); Chen et al. (2024); Zhang et al. (2024a); Marks and Tegmark (2023); Hoscilowicz et al. (2024)). The typical method for constructing a labeled CBQA dataset involves using a triplet (q, a_g, a_h), where q is a question, a_g is the gold answer, and a_h is the hallucinated answer. A hallucination example is created by concatenating a_h after q, while a factually-correct example is formed by appending a_g after q. This labeling approach is based on the correctness of the answers relative to world knowledge. However, many datasets only include the a_g answer, necessitating the creation of a_h. Following Li et al. (2023), we generate a_h by prompting an LLM (Mistral) to produce a plausible yet incorrect answer. See Appendix B for dataset creation details.

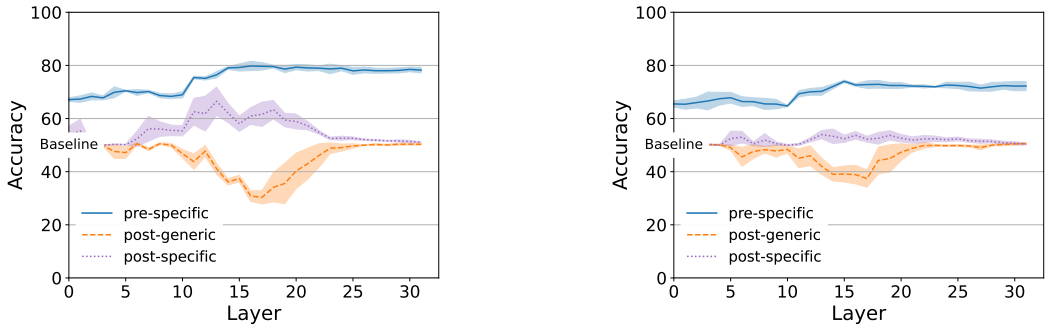
Results. As this work focuses on HK^+ cases (of hallucinations despite knowledge), we aim to show that the generic dataset is not effective in catching them. To this end, we compare classifiers trained in two binary settings: (1) model-specific setting, separating HK^+ from factually correct examples; and (2) generic setting, separating hallucinations from factually correct examples. We always test on the model-specific test set of HK^+ and factually correct examples. To make the generic and specific datasets more comparable, we add the Snowballing setting at the start of the prompt in both cases.

As Figure 5 shows, classifiers trained on hidden states obtained from generic datasets (dashed lines) demonstrate varying degrees of effectiveness and are always worse than classifiers trained on states from model-specific datasets (solid lines), which maintain relatively high accuracy. This comparison stresses the need to tailor hallucination detection to individual models, as it more effectively captures model-specific nuances and leads to more reliable identification of hallucinations per model. These results are consistent with related work (CH-Wang et al., 2023) that showed that a generic detector performs worse on specific datasets than one trained on specific datasets. However, unlike us, prior work used hallucination datasets that do not distinguish between the two hallucination types, HK^- and HK^+.



(a) Gemma. (b) Llama. (c) Mistral.

Figure 5: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on TriviaQA. Accuracy with classifiers trained on specific datasets is significantly higher than generic dataset. Appendix G has similar results on Natural Questions.



(a) TriviaQA (b) Natural Questions

Figure 6: Comparing preemptive HK^+ detection using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Mistral. Preemptive hallucination is effective using model-specific datasets, while detectors trained on generic datasets are ineffective.

6.2 Model-Specific Datasets Enable Preemptive HK^+ Detection

Our previous detection results used hidden states obtained after the model generated an answer, potentially already including a hallucination. A key advantage of model-specific datasets is their ability to detect potential hallucinations preemptively, *before they are generated*, a feature not possible with generic datasets. This section explores this capability using our WACK dataset (as before, with HK^+ and factually correct examples), where each example contains only the question q without an appended answer. As a result, the classifier is trained on the internal states of the examples at the last token of the question, rather than the last token of the answer.

Figure 6 displays preemptive hallucination detection results on the TriviaQA and Natural Questions datasets using the snowballing setting on Mistral. (See Appendix H for similar results on other models.) Model-specific preemptive detection (solid lines) demonstrates strong potential, suggesting that models can effectively anticipate hallucinations before they occur. In contrast, generic post-hallucination detection—evaluated at the last token of the answer (dashed lines)—shows random or even below random performance. This indicates that such an approach is ineffective in identifying HK^+ before their generation. Note that the generic approach can only be applied post-answer, as the hallucination signal depends on externally added annotations rather than originating from the model itself. In comparison, model-specific hallucination detection applied at the answer's last token (dotted lines), produces varied outcomes: for the TriviaQA dataset, some layers achieve detection rates of 60%–70%, while for Natural Questions, the detection rates remain low. We conclude that post-hallucination settings are not effective for preemptive hallucination detection, further highlighting the benefits of model-specific datasets.

7 Discussion and Conclusion

In this work, we demonstrated the prevalence of HK^+ across models, datasets, and settings. To induce HK^+, we developed a framework to create a labeled, model-specific dataset that categorizes outputs as HK^-, HK^+, or factually correct. We examined different configurations and observed some generalization between them, indicating that synthetic settings are effective for studying HK^+. Importantly, even mild conditions can induce HK^+, showing they manifest without flawed prompts. We highlighted the importance of distinguishing between hallucinations caused by a lack of knowledge (HK^-) and those occurring despite knowledge (HK^+) by showing that mitigation via prompting can only mit-